ASPIRE CODE AI

CODING THE FUTURE

INTERNSHIP PROJECT REPORT

Project Title: Superstore Sales Analysis Dashboard using PowerBI

Name / Intern ID	:	Abinaya E
Domain	:	Data Analytics
Duration	:	1 Month
Time Period	:	28-08-2026 To 28-09-2026
Supervisor / Mentor	:	Mr. Arunachalam M
TL Name	:	Mr. Arunachalam M

Submitted Date : 15/10/2026

Authority Signature Founder & CEO Aspire Code AI	**Report Incharge** COO & CTO Aspire Code AI

ABOUT ASPIRE CODE AI

Aspire Code AI is a next-generation technology startup specializing in Artificial Intelligence, Full-Stack Web Development, Business Analytics and Data Analytics. Our mission is to make AI-powered innovation accessible, scalable, and effective for businesses and educational institutions alike. Recognized under MSME India, we operate with professionalism, innovation, and a deep commitment to technological excellence.

VISION

To redefine the future of intelligent software by making AI-driven innovation accessible, scalable, and sustainable for every business.

MISSION

To deliver end-to-end AI and full-stack solutions that enhance productivity, automate workflows, and inspire creativity while maintaining transparency and excellence.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the course of this project.

First and foremost, I thank Mr. Arunachalam M, Founder & CEO of Aspire Code AI, for providing me with the opportunity to work on this project and for his visionary leadership that inspires innovation and learning.

I am deeply grateful to Ms. Vidurga Sabarisha J, COO & CTO, for her constant encouragement, guidance, and support during my internship period.

I would also like to thank Mr. Dineshkumar V, HR, for his valuable suggestions and for fostering an environment conducive to learning and experimentation.

My sincere thanks to Mr. Jabez Jebaraj D, ACTO, for his technical mentorship, innovative insights, and continuous support throughout the development of this project. I am equally grateful to Mr. Joel Godwin A, Technical Lead, for his guidance, technical expertise, and assistance in overcoming project challenges.

I would also like to extend my appreciation to Ms. Belicia Crescency, CCO, for her encouragement, professional guidance, and contributions toward enhancing my understanding of organizational communication and collaboration.

I would also like to thank Mr. Saravanakumaran M, CFO, for his guidance on aligning the project with organizational objectives and providing valuable perspectives that contributed to the successful completion of this work.

I am sincerely thankful to the entire Aspire Code AI team for their cooperation, assistance, and encouragement throughout the project.

Finally, I would like to extend my gratitude to my friends and family for their unwavering support, patience, and motivation, which were instrumental in the successful completion of this project.

Abinaya E
ACA/DAIS/0009/2026

DECLARATION

I hereby declare that this internship project report titled " Superstore Sales Analysis Dashboard using PowerBI " is a record of original work carried out by me during my internship at Aspire Code AI under the guidance of my mentor and team lead. This report has not been submitted, in full or in part, to any other organization or institution for any other purpose.

Place : KANYA KUMARI
Date : 15/10/2026

Signature of the Intern
Abinaya E
ACA/DAIS/0009/2026

CERTIFICATE

This is to certify that ABINAYA E (Intern ID: ACA/DAIS/0009/2026) has successfully completed the internship project titled " Superstore Sales Analysis Dashboard using PowerBI " in the domain of Data Analytics at Aspire Code AI for a duration of 1 month, from 28/08/2026 to 28/09/2026.

The work was carried out under the supervision of Mr. Arunachalam M and Mr. Arunachalam M. The intern showed dedication, discipline and a sound technical approach throughout the internship period.

Supervisor / Mentor Mr. Arunachalam M Aspire Code AI	**Team Lead** Mr. Arunachalam M Aspire Code AI
Report Incharge COO & CTO Aspire Code AI	**Authority Signature** Founder & CEO Aspire Code AI

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ABSTRACT

Write 150–250 words summarising the whole project: the problem, the approach, the technologies used, the outcome and the key learning. Write this last.

This project presents an interactive sales analysis dashboard built with Microsoft Power BI Desktop using the Sample Superstore dataset. Raw sales data stored in a CSV file was imported, cleaned and transformed using Power Query. A key issue was a date format mismatch: the file stored dates in month/day/year (US) format, while the system used day/month/year. This was resolved by converting the Order Date and Ship Date columns to Date type using the English (United States) locale. The cleaned data was then visualised through cards, bar charts, a line chart, a donut chart, a table of customers and interactive slicers for Segment and Region. The dashboard shows total sales of 2.30 million and total profit of 286.40 thousand, a profit margin of about 12.5%. It also shows that Technology and Furniture lead in sales and that the West and East regions contribute the largest share. The project strengthened my skills in data cleaning, data modelling and visual storytelling, and shows how a single interactive page can support business decision-making.

Keywords

Power BI, Data Analytics, Power Query, Dashboard, Sales Analysis, Data Visualisation

CHAPTER 1: INTRODUCTION

1.1 Background

Data analytics helps organisations turn raw data into decisions. Retail businesses generate large volumes of sales data, and dashboards make trends, profits and customer behaviour easy to see. Power BI is a widely used business intelligence tool for this purpose..

1.2 Problem Statement

Sales data in a raw CSV file is hard to read and analyse. Managers cannot easily see which categories, regions or customers drive sales and profit. Inconsistent data types and date formats also lead to wrong results if they are not corrected first.

1.3 Objectives

- *Import and clean the Superstore sales dataset using Power Query.*
- *Correct data type and date format issues.*
- *Build an interactive dashboard showing sales, profit, category, region, trend and customer performance.*
- *Provide slicers so users can filter data themselves.*

1.4 Scope of the Project

The project covers data import, cleaning, visualisation and interactive filtering in Power BI Desktop. It does not cover live data connections, automated refresh, predictive modelling or publishing to Power BI Service.

1.5 Organisation of the Report

Chapter 2 describes the internship. Chapter 3 reviews the tools and the proposed solution. Chapter 4 lists requirements. Chapter 5 explains the design and methodology. Chapter 6 covers implementation. Chapter 7 presents testing and results. Chapter 8 discusses challenges and learning. Chapter 9 gives the conclusion and future scope.

CHAPTER 2: INTERNSHIP OVERVIEW

2.1 Intern Details

Particulars	Details
Name	Abinaya E
Intern ID	ACA/DAIS/0009/2026
Domain	Data Analytics
Duration	1 Month
Time Period	28/08/2026 – 28/09/2026
Supervisor / Mentor	Mr. Arunachalam M
Team Lead	Mr. Arunachalam M

2.2 Roles and Responsibilities

List the tasks and responsibilities assigned to you during the internship.

- Importing and understanding the Superstore dataset.
- Cleaning and transforming data in Power Query.
- Designing and building dashboard visuals in Power BI.
- Testing slicer interactions and verifying numbers.
- Documenting the project and preparing the presentation.

2.3 Weekly Work Summary

Week	Dates	Activities / Tasks Completed	Outcome
Week 1	28/08/2026 – 03/09/2026	Learned the Power BI Desktop interface; explored the Superstore dataset and its columns	Understood the data structure
Week 2	04/09/2026 – 10/09/2026	Imported the CSV; cleaned data in Power Query; fixed the date locale issue (US format)	Clean dataset ready
Week 3	11/09/2026 – 17/09/2026	Built cards, bar charts, line chart, donut chart and customer table	Core dashboard ready
Week 4	18/09/2026 – 24/09/2026	Added Segment and Region slicers; arranged layout; tested filters and verified figures	Interactive dashboard completed
Week 5	25/09/2026 – 28/09/2026	Final documentation, project report and presentation preparation	Final dashboard and report submitted

Add or remove rows to match your internship duration.

CHAPTER 3: LITERATURE / TECHNOLOGY REVIEW

3.1 Existing Systems / Related Work

Businesses traditionally analyse sales in spreadsheets such as Excel. These are manual, hard to filter interactively, and become slow and error-prone with large data.

3.2 Proposed Solution

A Power BI dashboard loads the data once, cleans it with repeatable Power Query steps, and presents all key metrics on one interactive page. Slicers let users filter instantly without editing formulas..

3.3 Technologies and Tools Used

Category	Technology / Tool	Purpose
Programming Language	M language (auto-generated by Power Query); DAX (for measures)	M handles data cleaning steps such as type conversion. DAX is used for calculated measures like Profit Margin
Framework / Library	Power BI Desktop built-in visuals	Cards, bar charts, line chart, donut chart, table and slicers for the dashboard
Database	Sample - Superstore.csv; Power BI data model	Source dataset (21 columns) and the in-memory model the visuals read from
AI / ML Tools	Not applicable	No AI/ML model was used in this project
Version Control	Not used (local .pbix file)	Dashboard saved as Superstore Dashboard.pbix
Other Tools	Power Query Editor, MS Word, MS PowerPoint	Data cleaning, report writing and presentation

CHAPTER 4: SYSTEM ANALYSIS AND REQUIREMENTS

4.1 Functional Requirements

- Import CSV data into Power BI.
- Convert Order Date and Ship Date to Date type correctly.
- Display total sales and total profit.
- Show sales by category, profit by sub-category and sales by region.
- Show the monthly sales trend and the customer table.
- Filter all visuals using Segment and Region slicers.

4.2 Non-Functional Requirements

Accuracy of figures, quick response when filtering, a clear and readable layout, and easy use by non-technical users.

4.3 Hardware Requirements

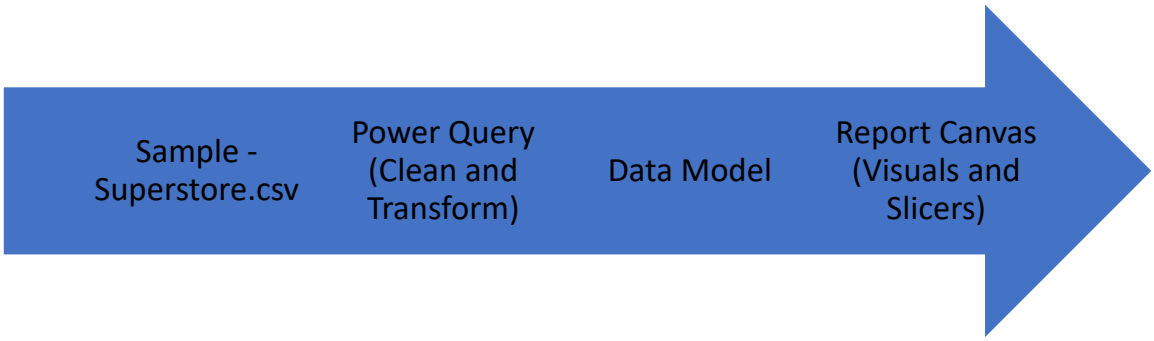
Component	Specification
Processor	12th Gen Intel(R) Core(TM) i5-12450H (2.00 GHz)
RAM	16.0 GB (15.7 GB usable)
Storage	165 GB of 477 GB used
Other	Laptop display (15.6"), keyboard, mouse, internet connection (Wi-Fi)

4.4 Software Requirements

Software	Version / Details
Operating System	26200.9711
IDE / Editor	2.147.1085.0 64-bit
Runtime / Compiler	Not applicable (no-code tool)
Other	Power Query Editor, MS Word, MS PowerPoint, Google Chrome

CHAPTER 5: SYSTEM DESIGN AND METHODOLOGY

5.1 System Architecture



5.2 Workflow / Data Flow Diagram

Import → Transform Data → Fix data types and date locale → Remove unneeded columns → Close & Apply → Build visuals → Add slicers → Test → Save.

5.3 Database Design / Data Model

Single table, "Sample - Superstore", with 21 columns. Key columns: Order Date, Ship Date, Ship Mode, Customer Name, Segment, Country, State, City, Region, Category, Sub-Category, Sales, Quantity, Discount, Profit.

5.4 Methodology / Algorithm

1. *Opened Get Data → Text/CSV and checked the preview (Delimiter: Comma).*
2. *Chose Transform Data instead of Load to clean the data first.*
3. *Found that Order Date and Ship Date were Text type.*
4. *Converted them using Change Type → Using Locale → Date, English (United States), because the file uses month/day/year format.*
5. *Removed the Row ID column and checked the Sales, Profit, Discount and Quantity types.*
6. *Applied changes with Close & Apply and built the visuals.*

CHAPTER 6: IMPLEMENTATION

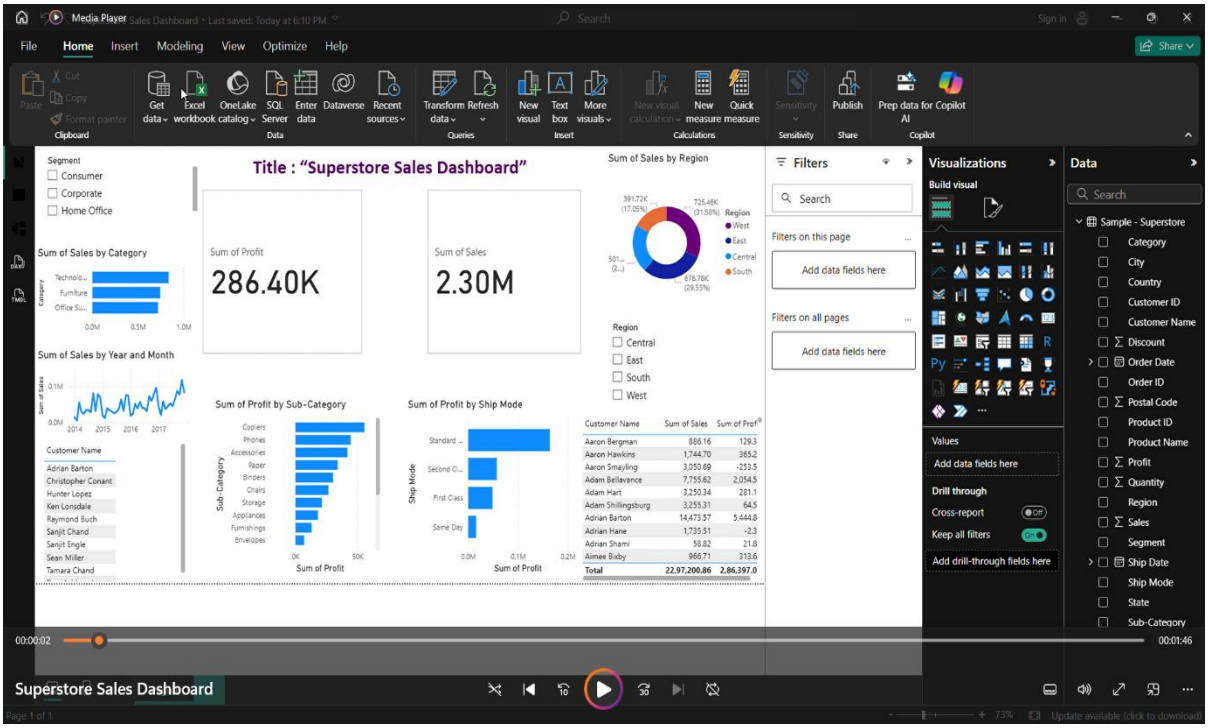
6.1 Module Description

Module	Description
Module 1	Loads the CSV into Power BI
Module 2	Fixes types, date locale and removes unused columns
Module 3	Cards, bar charts, line chart, donut chart, table
Module 4	Segment and Region slicers filter all visuals

6.2 Key Code Snippets

The project is mostly no-code. The M step generated by Power Query for the date fix:
Table. Transform Column Types(#"Promoted Headers", {{"Order Date", type date}, {"Ship Date", type date}}, "EN -US")

6.3 User Interface / Output Screenshots



CHAPTER 7: TESTING AND RESULTS

7.1 Testing Approach

Manual testing of data types, slicer filtering and figure consistency.

7.2 Test Cases

ID	Test Case	Expected Result	Actual Result	Status
TC01	Date conversion (EN - US locale)	Error 0%, Valid 100%	Error 0%, Valid 100%	Pass
TC02	Row 1 Order Date 11/8/2016	Shows 8 Nov 2016	Shows 08-11-2016	Pass
TC03	Select Consumer and South in slicers	Visuals filter	Sales card showed 195.58K	Pass
TC04	Clear slicers	Totals return to full	Sales card back to 2.30M	Pass

7.3 Results and Analysis

- *Total Sales: 2.30 million. Total Profit: 286.40 thousand. Profit margin: about 12.5%.*
- *Technology and Furniture lead in category sales.*
- *West (31.58%) and East (29.55%) contribute the largest regional shares.*
- *The monthly trend runs from 2014 to 2017 and shows a seasonal pattern.*
- *Some sub-categories have low profit, so they need attention.*

CHAPTER 8: CHALLENGES AND LEARNING OUTCOMES

8.1 Challenges Faced and Solutions

Challenge	Solution Adopted
Dates imported as Text	Converted to Date type
Date format mismatch (US vs system format) gave wrong dates and errors	Used Change Type → Using Locale → English (United States)
Clicked the wrong visual (Q&A)	Deleted it and chose the correct icon
Map visual disabled by security setting	Map visual disabled by security setting

8.2 Technical Skills Gained

Power BI Desktop, Power Query, data type and locale handling, visual selection, slicer interactivity, dashboard layout.

8.3 Professional / Soft Skills Gained

Problem-solving, attention to detail, time management (completed within a deadline), and learning independently.

8.4 Contribution to Organisation

Created a reusable dashboard template and a documented data-cleaning approach that can be applied to other sales datasets.

CHAPTER 9: CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

The objectives were met. Raw data was cleaned and turned into an interactive dashboard that gives clear sales and profit insights on one page.

9.2 Limitations

Static CSV source with no auto-refresh, no map visual, no forecasting, and the dashboard is not yet published online.

9.3 Future Scope

- Add a map visual for state-wise sales.
- Add a Profit Margin measure using DAX.
- Add a Year slicer and red conditional colours for loss-making sub-categories.
- Publish to Power BI Service and schedule refresh.

REFERENCES

List all books, papers, documentation and websites used (IEEE or APA style).

[1] Microsoft, “Power BI documentation,” Microsoft Learn, https://learn.microsoft.com/power-bi/
[2] Microsoft, “Power Query documentation,” Microsoft Learn, https://learn.microsoft.com/power-query/
[3] Sample - Superstore dataset (Tableau sample data).

APPENDICES

Appendix A: Complete Source Code

Appendix A: Complete Source Code

This project was developed using Microsoft Power BI Desktop following a no-code approach. The dashboard was created using Power Query for data cleaning and transformation, along with Power BI visualizations, calculated measures, and interactive filters.

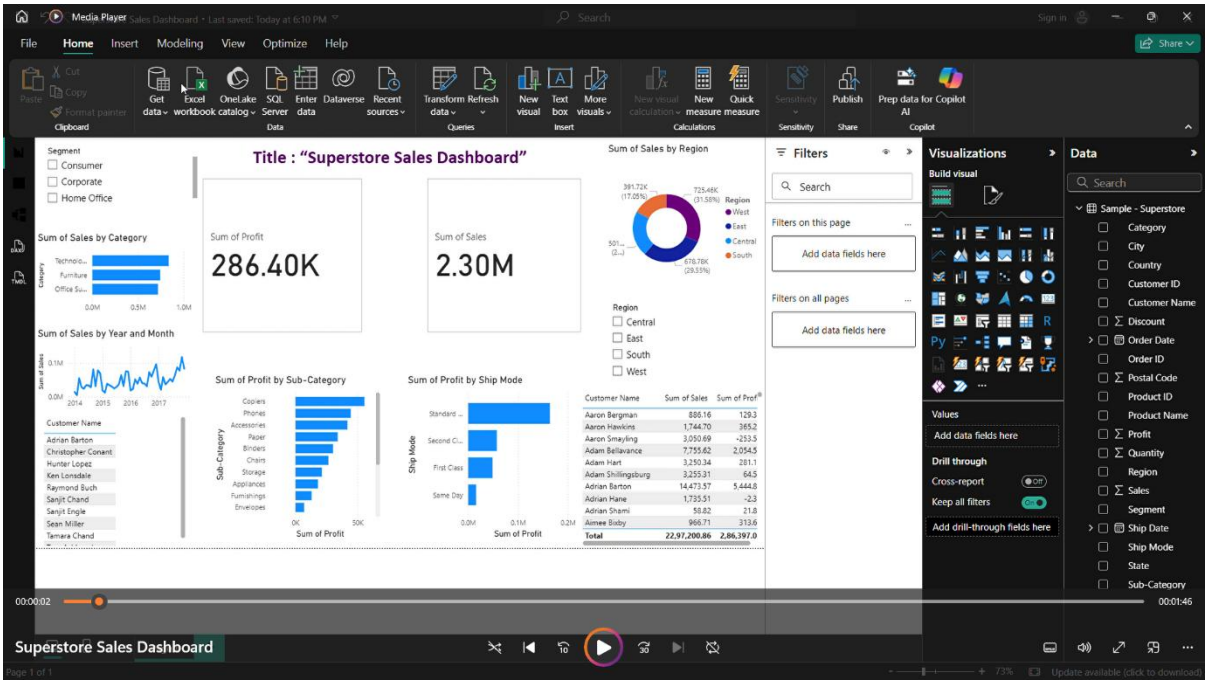
The following Power Query M code was used to convert the *Order Date* and *Ship Date* columns into the date data type using the US English locale.

Power Query (M) Code:

```
Table. Transform Column Types (
   #"Promoted Headers",
   {
      {"Order Date", type date},
      {"Ship Date", type date}
   },
   "EN-US"
)
```

The complete dashboard and its associated data transformations are saved in the Power BI project file, **Superstore Dashboard. pbix**. This file contains the dashboard visualizations, data model, measures, and report design used for the Superstore Sales and Profit Analysis project.

Appendix B:



Appendix C: Project Links

Item	Link
GitHub Repository	https://github.com/abinaya21offl-droid/Superstore-Sales-Dashboard-Power-BI/upload
Live Demo	Not Applicable
Documentation	This report
Demo Video	Not Applicable

Appendix D: Mentor Feedback

Mentor / Team Lead remarks on performance and conduct.

Evaluation Parameter	Rating (out of 10)	Remarks
Technical Knowledge		
Problem Solving		
Punctuality & Discipline		
Teamwork & Communication		
Project Completion		